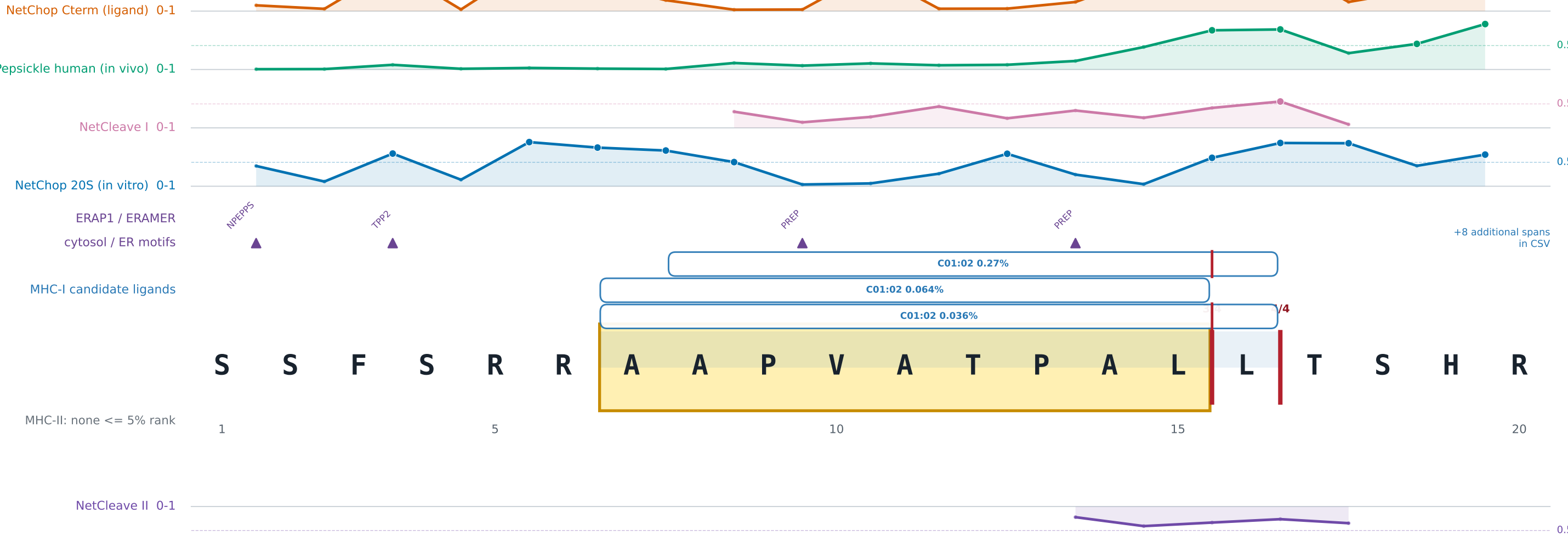


PIP5K1A p.Gly474fs | JLF V3 | 20 aa

disclosed minimal epitope AAPVATPAL at 7-15

INTRACELLULAR / CLASS-I PROCESSING - four complementary score tracks



SERUM / EXTRACELLULAR - native scores and recognition motifs



SCORES Native 0-1 bond scores; lines join adjacent assessed bonds only (no smoothing). Dashed line = 0.5 display threshold; gaps = unassessed.

RED CUTS All four intracellular tracks must assess the bond and at least three must score >=0.5. The 3/4 or 4/4 label is support, not probability.

MHC WINDOWS Blue/purple tint and outlines show displayed <=2%/<=5% rank candidates; internal red ticks are pre-binding cut evidence. All candidates remain in CSV.